Agile Methodology (Definition and Basics):

Agile Methodology is a product development lifecycle framework, predominantly used in software development, which emphasizes iterative development, incremental delivery, customer collaboration, continuous feedback, and responding to changing requirements.

Derived from the Agile Manifesto of 2001, it focuses on 5 C's as core principles, *Communication, Collaboration, Commitment, Customer and Continuous Improvement* to harbour a setting for open communication, engaging teamwork and continuous focused improvement to deliver the best product for customer satisfaction.

Typical Agile framework involves:

1. **Planning:** Defining goals and requirement based on customer needs, thus planning further work
2. **Development:** Collaborative development to produce selected functionalities
3. **Testing/Validation:** Testing the product via comparing with requirements declared before, finding defects and solving them ahead of delivering
4. **Incremental Delivery:** Delivering the product in increments, taking feedback from stakeholders in due process and improving the overall quality of both the product and process for maximum stakeholder satisfaction.
5. **Review and Feedback:** Evaluating the work and taking feedback from stakeholders and client to guide further development
6. **Adaptation and Improvement**: Adapting the process, requirements or priorities based on feedback from stakeholders.

Some of the most used Agile Frameworks are Scrum, Kanban, Extreme Programming(XP) which are mostly used in software development.

Scrum:

Scrum is a widely used agile management framework that helps teams organize and oversee their work through collaboration over iterative work. It emphasizes an iterative and incremental approach, thus helping teams to self organize and collaborate to deliver High quality software even if the product is complex.

Scrum follows the agile framework and Refines it, makes it more executable. Scrum uses sprints, which last from two to four weeks, depending on the requirements.

Scrum follows a 3-5-3 rule that defines the roles, framework and requirements (named as artifacts) of scrum. These are,

3 Roles:

1. **Product owner**: They are responsible for understanding the business, customer and market requirements. They communicate the product goal and communicate the product backlogs with the rest of the stakeholders. They represent the customer.

2. **Scrum Master:** Are responsible for coaching teams on self management and collaboration, product owners on effective goal declaration and communication, and the business by leading on the Scrum process and look for ways to fine-tune their practice.
3. **Development Team:** The team that designs and develops the product.

Scrum Teams are always to be a small, nimble yet tight-knit group roughly around 10 people.

5 Events:

1. **Sprint:** A short time boxed period that the scrum team uses to complete a set amount of work. It can be defined as a short project within the domain of a larger project. It must have clear goals, a defined body of work for proper planning,execution, review and later on improvements.
2. **Sprint Planning:** The first event of scrum framework and subsequently also a sprint, A sprint planning defines the work that needs to be done within a sprint. It connects product backlog to daily team work.
3. **Daily Scrum:** Also known as daily Standups or sitdown, it focuses on the development team conveying their progress, todos and blockers to the product owner or scrum master
4. **Sprint Review:** A collaborative working session held at the end of a sprint where the Scrum Team and key stakeholders inspect the product increment, demonstrate completed work, and adapt the Product Backlog
5. **Sprint Retrospective:** An event where the scrum team inspects how the last Sprint went with regards to individuals, interactions, processes, tools, and their Definition of Done.

3 Artifacts:

1. **Product Backlog:** Product Backlog is the primary dynamic list of features, requirements, enhancements, and fixes that the product owner or product manager needs to do and maintain. It dictates the sprint backlog.
2. **Sprint Backlog:** Sprint Backlog is the list of items, user stories, or bug fixes, selected by the development team from the product backlog for implementation in the current sprint cycle.
3. **Increment/Definition of Doneness:** Defines how you define your sprint goals as done, defers organization to organization, Example: some may say release of a sprint is increment others may say testing and review as a definition of doneness.

Resources:

1. **https://www.atlassian.com/agile**
2. **https://www.atlassian.com/agile/scrum**
3. **https://www.geeksforgeeks.org/software-testing/what-is-agile-methodology/**
4. **https://www.geeksforgeeks.org/software-engineering/scrum-software-development/**
5. **https://www.scrum.org/resources/what-is-a-product-owner**
6. **https://www.scrum.org/resources/what-is-a-scrum-master**
7. **https://www.scrum.org/resources/what-is-a-sprint-retrospective**
8.